



ADDIS ABABA SCINENCE AND TECHNOLOGY UNIVERSITY
COLLEGE OF SOFTWARE ENGINEERING
SWEG3103 - Data Structure and Algorithms

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1 Objective

The main goal of this activity was to understand how a Queue (FIFO) works by actually acting it out with people. Instead of just reading about it in the textbook, we physically stood in a line and performed the operations. This helped us see step by step how data moves in and out of a queue, and we also got to practice counting operations and managing pointers. This experience will help us with our bigger DSA project later.

2 .Why We Chose Queue

We picked Queue (FIFO) from Chapter 4 because:

- It's easy to visualize – a line of people is something everyone understands.
- It has clear operations – Enqueue (adding to the back) and Dequeue (removing from the front) are simple to act out.
- It's useful – we know queues are used in many real systems like printer queues, call centers, and even in graph algorithms like BFS.
- It's a good challenge – even though it seems simple, managing the front and rear pointers and keeping count of operations was interesting for our group of 7.

3 .Group Roles

We divided the work among our 7 members so everyone had something to do:

Member	Role	What They Did
1 Amanuel	Narrator	Announced each operation, told people where to move, explained things in the video
2.Paulos	Node 1	Stood at the front with card "10"
3 Michael	Node 2	Stood in the middle with card "25"
4. Leul	Node 3	Stood at the back with card "5"
5 Mulusew	Node 4	Joined later with card "40"
6 Michael	Node 5	Joined later with card "15"

7. wubzema	Scribe / Camera	Kept count of operations and recorded the video
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4 Materials We Used

We didn't use anything fancy – just simple things we had around:

Item	What It Was For
Numbered cards (10, 25, 5, 40, 15)	Each person held one to show their "data"
Orange cone	Marked who was at the front
Blue cone	Marked who was at the back
"EMPTY" card	Showed when the queue was empty
Notebook	Scribe used it to write down operation counts
Smartphone	We recorded the whole simulation

5.How We Started

We began with three people already in the queue so we could show different operations:

FRONT $\rightarrow$ [10] $\rightarrow$ [25] $\rightarrow$ [5] $\leftarrow$ REAR

- Node 1 (10) was at the front
- Node 2 (25) was in the middle
- Node 3 (5) was at the back

This setup let us show Enqueue (adding more people), Dequeue (removing people), Peek (checking the front), and what happens when the queue becomes empty.

6 What We Did – Step by Step

Here's everything we did in order, with our tally:

Step	Operation	What Happened	Queue After	Count
1	Start	Three people stood in line	[10] $\rightarrow$ [25] $\rightarrow$ [5]	

2	Enqueue(40)	Node 4 joined at the back	[10] → [25] → [5] → [40]	Enq: 1
3	Enqueue(15)	Node 5 joined at the back	[10] → [25] → [5] → [40] → [15]	Enq: 2
4	Peek()	Node 1 showed card "10"	No change	
5	Dequeue()	Node 1 stepped out	[25] → [5] → [40] → [15]	Deq: 1
6	Dequeue()	Node 2 stepped out	[5] → [40] → [15]	Deq: 2
7	Enqueue(30)	Node 6 joined at the back	[5] → [40] → [15] → [30]	Enq: 3
8	Dequeue()	Node 3 stepped out	[40] → [15] → [30]	Deq: 3
9	Dequeue()	Node 4 stepped out	[15] → [30]	Deq: 4
10	Dequeue()	Node 5 stepped out	[30]	Deq: 5
11	Dequeue()	Node 6 stepped out	[EMPTY]	Deq: 6
12	Empty / Underflow	Showed "EMPTY" card and explained underflow	[EMPTY]	-

7. Count of Operations

Here's the final tally of everything we did

Operation	How Many
Enqueue	3
Dequeue	6
Peek	1
Empty shown	1
Underflow attempt	1

8 .Challenges We Faced

Even though the simulation was helpful, we ran into some problems:

1. Keeping the Line Order Right

In theory, FIFO is simple. But in practice, we had to make sure nobody accidentally cut in line or stood in the wrong place. The narrator had to keep watching and telling people where to go. This showed us why pointers need to be managed carefully in real code.

2. Moving Pointers Was Clunky

Using cones to show the front and rear was okay, but it was slow. Every time we did an operation, someone had to physically move a cone. In a real program, updating a pointer happens instantly. We couldn't really show how fast these operations are.

3 What to Do With People Who Left

When someone did Dequeue, they stepped out of line and just stood there waiting. In a real system, the memory is freed or reused. We had to figure out where to put the people who left, which was an awkward physical problem that doesn't exist in code.

4. Overflow Was Hard to Show

Since we only had 7 people, we couldn't really show what happens when a queue gets full (overflow). We could have used a sign, but it's not the same as actually experiencing it. This shows the difference between a fixed-size array queue and a conceptual queue.

9. Video and GitHub Links

We recorded a 5-minute video showing the whole simulation with narration. You can see all the operations we did, with the narrator explaining each step.

Video Link:

Our project files, including this report, are on GitHub:

https://drive.google.com/file/d/1RTC-r7IZytrOu3AQWeyqf4fmzAMCmAaH/view?usp=drive_link
GitHub: [https://github.com/\[your-username\]/DSA-Roleplay-Ch2-6](https://github.com/[your-username]/DSA-Roleplay-Ch2-6)

10 Conclusion

This activity helped us understand queues much better than just reading about them. Actually being the nodes and moving around made the FIFO principle really clear.

The main things we learned:

- FIFO means first in, first out – the person who joins first leaves first
- Managing front and rear pointers is important and needs to be done carefully
- Counting operations helps understand how efficient algorithms are
- Error handling (like underflow) is something real systems have to deal with

Everyone in the group participated and we all feel more confident about the topic now. This was a good preparation for our main project.

References

1. SWEG3103 Course Material – Chapter 4 (Queue)
2. Project instruction document – DSA Role-Play Simulation Guidelines